

Final Project Dataset Overview: Suspicious Tax Returns

Introduction

This dataset is designed to help you analyze patterns in tax filings, understand IRS adjustment behaviors, and develop predictive models to estimate the likelihood and extent of tax return adjustments.

Dataset Description

Each row represents an individual tax return, with various financial and demographic variables. The goal is to build a regression model that predicts Tax_Adjustment_Amount, which reflects the IRS’s corrections to the reported tax filings.

Response Variable

- **Tax_Adjustment_Amount** (*continuous variable*)
 - The total dollar amount by which the IRS adjusted the taxpayer’s reported return.
 - Influenced by income, deductions, business expenses, and IRS scrutiny patterns.

Predictor Variables

Variable	Description
Reported_Income (\$)	Total income reported on the tax return.
Claimed_Deductions_Pct (%)	Claimed deductions as a percentage of income.
Business_Expenses (\$)	Total reported business expenses (for self-employed and small business owners).
Charitable_Contributions_Pct (%)	Charitable donations as a percentage of income.
Home_Office_Deduction (\$)	Amount claimed for home office expenses.
Num_Dependents	Number of dependents reported.
Amended>Returns	Number of times the taxpayer has submitted amended returns.
Tax_Credits (\$)	Total claimed tax credits (e.g., child tax credit, education credit).
Investment_Income_Pct (%)	Percentage of total income that comes from investments (stocks, dividends, etc.).
Income_Variability_Pct (%)	Year-over-year percentage change in reported income.
Years_Since_Last_Audit	Number of years since the taxpayer’s last IRS audit.
Occupation	Categorical variable representing the taxpayer’s job.

Occupation Categories

The dataset includes five different occupations, each affecting IRS scrutiny differently:

- **Self-Employed** (e.g., freelancers, gig workers)
- **Small Business Owner** (e.g., shop owners, consultants)
- **Doctor** (lower risk of audit due to structured financials)
- **Lawyer** (lower risk, detailed record-keeping)
- **Real Estate Agent** (higher risk, large deductions)

This dataset is designed to help you think critically about financial fraud detection, tax behavior, and statistical modeling. Use this opportunity to explore realistic tax-related decision-making and build a compelling analysis!

Hints:

1. To get a good model, you will want adjusted R-squared to be above 0.8.
2. You may want to create the dummy variables manually.
3. It is suspected that Small Business Owner returns in this dataset behave very differently than the rest.